SKILLSWAP

A Community-Driven Skill Barter Platform

A Real-Time Research Project / Societal Related Project submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

By

GOPU HARDIK	**(24281A6710)**
MUDDAM SREEJA	**(24281A6715)**
KOLLURI SRIGANDHA	**(24281A6727)**
MD. MEHTAB ALI	**(25285A6701)**

PROJECT GUIDE

Dr. K. Praveen Kumar Rao

HOD & Professor, Department of CSE (DS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

KAMALA INSTITUTE OF TECHNOLOGY & SCIENCE

(UGC AUTONOMOUS)

(Sponsored by Kamala Education Society, Approved by AICTE, New Delhi, Affiliated to JNTUH, Hyderabad, Telangana, Accredited by NBA (CSE, ECE & EEE) and Accredited by NAAC with A++ Grade)

SINGAPUR, HUZURABAD, KARIMNAGAR, TELANGANA – 505 468

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C E R T I F I C A T E

This is to certify that **GOPU HARDIK (24281A6710), MUDDAM SREEJA (24281A6715), KOLLURI SRIGANDHA (24281A6727), MD. MEHTAB ALI (25285A6701)** students of the II B. Tech – II Semester Computer Science and Engineering (Data Science) have satisfactorily completed their ***Real-Time Research Project / Societal Related Project*** entitled **"SKILLSWAP – A Community-Driven Skill Barter Platform"** towards the partial fulfilment of B. Tech degree during the academic year 2025–26.

Project Guide **Head of the Department**

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Gopu Hardik	(24281A6710)
Muddam Sreeja	(24281A6715)
Kolluri Srigandha	(24281A6727)
MD. Mehtab Ali	(25285A6701)

ABSTRACT

SkillSwap is a Real-Time Resource Pool (RTRP) — a web-based platform designed to democratize education by replacing monetary costs with a skill-barter system. Addressing financial barriers to upskilling, it connects individuals who wish to teach and learn, making knowledge the currency of exchange. The platform operates on a time-banking concept where one hour of teaching equals one hour of learning credit, enabling direct peer-to-peer knowledge exchange without monetary transactions.

The platform features a responsive React.js frontend powered by Vite and styled with Tailwind CSS, alongside a Node.js and Express.js backend providing RESTful API architecture. The application supports secure user registration and login using JWT (JSON Web Tokens) with bcrypt password hashing. Key capabilities include comprehensive profile management, intelligent skill discovery with category-based filtering, a real-time messaging system, and online status tracking.

The system is orchestrated around four core modules: User Management, ensuring secure authentication and profile verification; Skill Discovery, offering advanced filtering for precise skill matching; Skill Management, allowing users to curate their expertise portfolio; and a Communication System, facilitating real-time negotiation and scheduling of learning sessions.

SkillSwap directly addresses key problems in modern education — high tutoring costs, one-way learning, undervalued peer skills, and the lack of a structured community for mutual growth. The platform was successfully developed and tested, demonstrating the viability of peer-to-peer learning through time banking, and providing a strong foundation for future expansion into a production-ready educational platform.

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CHAPTER 1

INTRODUCTION

1.1 About the Project

In the modern knowledge economy, access to quality education and skill development remains a significant challenge for students, professionals, and hobbyists alike. While digital platforms have expanded access to learning, they continue to rely on monetary transactions that create barriers for those with limited financial resources. Peer-to-peer learning — where individuals teach what they know and learn what they need — has proven to be one of the most effective and economical approaches to education, yet no dedicated, modern digital platform exists to facilitate this exchange in a structured manner.

SkillSwap is a Real-Time Resource Pool (RTRP) — a web-based platform built to bridge this gap. It enables peer-to-peer skill exchange using a time-banking model, where one hour of teaching earns one hour of learning credit. This approach replaces money with time as the medium of exchange, making quality learning accessible to everyone regardless of financial background.

The platform allows users to register, create a profile showcasing their skills and expertise, browse skills offered by community members, and connect directly through a built-in messaging system. The application is developed using React.js on the frontend with Vite and Tailwind CSS, and Node.js with Express.js on the backend, ensuring a modern, responsive, and scalable full-stack web application.

SkillSwap aligns with the broader objective of democratizing education and building a self-sustaining learning community where knowledge is shared freely and equitably. The project demonstrates the practical application of full-stack web development skills to solve a real-world educational and social problem.

1.2 Existing System with Drawbacks

Currently, skill development and learning opportunities are provided through commercial platforms and informal methods. While these systems have contributed significantly to online education, they suffer from limitations that restrict access for a large portion of the population.

Platforms such as Coursera, Udemy, and Skillshare operate on subscription or pay-per-course models, making them inaccessible to learners with limited budgets. The learning model is predominantly one-directional — learners consume pre-recorded content without interactive feedback or personal mentorship. Freelance marketplaces like Upwork and Fiverr are entirely transactional, providing no mechanism for skill exchange or community-driven learning. Traditional time-banking communities, while conceptually aligned with SkillSwap, remain largely offline and localized, lacking the digital infrastructure required for scalable adoption.

Drawbacks of Existing Systems:

- High cost of professional tutoring and course subscriptions limits accessibility
- One-way, passive learning with no direct peer interaction or feedback
- Undervalued skills — many individuals have valuable expertise but no platform to share it for equivalent return
- Lack of a dedicated community space for structured peer-to-peer skill exchange
- Traditional time banks are offline, unscalable, and not equipped with modern communication tools
- No integrated platform combining skill discovery, profile management, and real-time communication

1.3 Proposed System with Features

The proposed system, SkillSwap, is a full-stack web application that modernizes the concept of time banking for skill exchange. It provides a unified platform where users can both offer their skills and seek skills they wish to learn, all without monetary transactions. The system is built using React.js, Node.js, Express.js, and MySQL, with JWT-based authentication ensuring security.

SkillSwap enables users to register and create comprehensive profiles detailing their expertise, interests, bio, location, and time-credit balance. Each user starts with 10 time credits. A centralized skill marketplace allows users to browse, filter, and discover skills by category. A real-time messaging system facilitates direct communication between learners and teachers to negotiate sessions and coordinate learning activities.

Features of the Proposed System:

- Time-based economy: 1 hour of teaching = 1 hour of learning credit
- Peer-to-peer connection: users find and connect based on specific skills
- Skill Marketplace: browse and filter skills by category
- Secure user authentication using JWT and bcrypt password hashing
- Real-time messaging system for direct communication between users
- Comprehensive user profiles with bio, role, location, and time-credit balance
- Online status tracking to identify available users
- Dashboard with community statistics, featured skills, and member discovery
- Responsive, modern UI built with React.js, Vite, and Tailwind CSS

CHAPTER 2

LITERATURE SURVEY

The concept of skill exchange and time-based economy has been explored academically and practically in various forms. This chapter presents a survey of existing systems and research related to peer-to-peer learning platforms, time banking, and skill marketplaces. The purpose of this survey is to understand the current landscape, identify limitations, and justify the need for the proposed SkillSwap platform. The survey covers a maximum of three pages as required.

2.1 Study of Existing Peer-to-Peer Learning Platforms

Several commercial and community-driven platforms have attempted to address the problem of accessible education. Coursera and Udemy pioneered online learning by aggregating expert-created content into structured courses. However, these platforms are primarily consumption-based — learners purchase courses and watch pre-recorded videos without direct interaction with instructors or peers. The monetary barrier remains significant, particularly for learners in developing economies.

Platforms like Wyzant and Preply provide on-demand tutoring and allow learners to connect with tutors directly, but these are entirely money-driven marketplaces. Tutors charge hourly fees, making sustained learning expensive. Skillshare introduced subscription-based community projects, encouraging slightly more interaction, but still depends on content creators and does not facilitate direct skill exchange between ordinary users.

2.2 Time Banking and Skill Exchange Concepts

Time banking was originally proposed by Edgar Cahn in the 1980s as an alternative economic model where time, rather than money, is the unit of exchange. In a time bank, one hour of any service is equal to one hour of any other service, regardless of the perceived market value. This concept has been applied in community service networks worldwide but has rarely been translated into a scalable digital skill exchange platform.

Research studies on time banking communities have consistently shown that they improve social capital, reduce feelings of isolation, and provide access to skills and

services for economically disadvantaged populations. However, the lack of modern digital infrastructure has limited their scalability and reach.

2.3 Gaps in Existing Systems and Need for SkillSwap

After studying various existing platforms and research, the following key gaps were identified:

- Monetary dependency: All major platforms require financial payment, creating barriers for resource-constrained learners
- Passive learning: Pre-recorded content platforms lack direct, interactive, personalized experiences
- No structured barter mechanism: No existing digital platform provides a structured skill-for-skill exchange with formal credit tracking
- Poor community integration: Most platforms are transactional rather than relational — they do not foster genuine communities of mutual learning
- Offline limitations of traditional time banks: Existing time-banking communities lack digital tools for communication, skill discovery, and session management

The literature survey conclusively demonstrates that while numerous platforms address individual aspects of the problem, none provides an integrated solution combining skill discovery, peer-to-peer exchange, time-credit management, and real-time communication in a single accessible digital platform. SkillSwap is designed to address all these gaps directly.

CHAPTER 3

SYSTEM ANALYSIS

System analysis is the process of studying the proposed system to understand its requirements and functionality. In this phase, the hardware, software, functional, and non-functional requirements of the SkillSwap application are analyzed in detail to ensure a robust, secure, and user-friendly platform is designed and developed.

3.1 Hardware and Software Requirements

Table 3.1 – Hardware Requirements

Processor	Intel Core i5 (8th Generation or above)
RAM	8 GB or more
Storage	256 GB SSD or higher
Display	Any standard LED/LCD monitor
Input Devices	Standard keyboard and mouse
Internet Connection	Broadband or Wi-Fi connection

Table 3.2 – Software Requirements

Operating System	Windows 10 / MacOS / Ubuntu (any modern OS)
Frontend Framework	React 19 with Vite
CSS Framework	Tailwind CSS
Backend Runtime	Node.js (v18 or above)
Backend Framework	Express.js
Database	MySQL with mysql2 driver
Authentication	JWT (JSON Web Tokens) + bcrypt
IDE / Editor	Visual Studio Code
API Testing Tool	Postman

Version Control	Git and GitHub
Web Browser	Google Chrome / Mozilla Firefox / Microsoft Edge
Deployment Platform	Vercel (Frontend)

3.2 Functional Requirements

Functional requirements describe the core operations and features that the SkillSwap system must perform:

1. User Registration and Authentication: The system shall allow new users to register using a unique username, email, and password. Passwords must be hashed using bcrypt before storage, and session management must use JWT tokens with a 1-hour expiry.
2. User Profile Management: Each user shall have a profile containing their name, email, bio, role, location, website URL, time-credit balance, and avatar generated via DiceBear API. Users should be able to edit and update their profile information.
3. Skill Posting and Management: Users shall be able to create skill listings with a title, description, category, and estimated hours. Users should be able to edit and delete their own listings.
4. Skill Discovery and Browsing: The system shall provide a marketplace where all listed skills are displayed. Users shall be able to filter skills by category and search for specific skills.
5. Messaging and Communication: Users shall be able to send and receive persistent messages to/from other users, with full message history maintained per conversation.
6. Online Status Tracking: The system shall track and display the real-time online/offline status of users.
7. Dashboard: The platform shall display community statistics including total registered members, skills listed, and currently online users, along with a featured skills grid and member discovery section.

8. Time Credit System: Each user shall start with 10 time credits. Credits are earned by teaching and spent for learning sessions.

3.3 Non-Functional Requirements

1. Performance: The application should load within 2 seconds on a standard broadband connection. API responses should return within 500 milliseconds for standard queries.
2. Security: All passwords must be stored as bcrypt hashes. JWT tokens expire after 1 hour. All sensitive API routes are protected with authentication middleware. SQL injection is prevented through parameterized queries.
3. Usability: The user interface must be responsive and work correctly on both desktop and mobile browsers. Navigation must be intuitive and require no prior technical knowledge.
4. Scalability: The system architecture shall support future additions such as video calling, a rating system, AI matching, and a notification engine without major refactoring.
5. Reliability: The application must handle invalid inputs gracefully and display meaningful error messages to users.
6. Maintainability: Code must follow a modular structure with clear separation of frontend components, backend routes, middleware, and database queries.

3.4 Module Description

The SkillSwap application is divided into four primary modules:

1. User Management Module: Handles user registration, login, logout, and profile management. Includes JWT token generation and validation, bcrypt password hashing, and online status updates.
3. Skill Management Module: Allows users to create, view, edit, and delete skill listings. Manages the marketplace including browsing and filtering by category.
4. Communication Module: Manages one-to-one messaging between users. Stores and retrieves message history from MySQL. Handles real-time online status updates.
5. Dashboard and Discovery Module: Displays community statistics. Provides a featured skills grid and member discovery section for exploration and navigation.

CHAPTER 4

SYSTEM DESIGN

System design translates the requirements identified in the analysis phase into a structured blueprint for the SkillSwap application. This chapter covers the block diagram, data flow diagrams, ER diagram, UML diagrams, and data dictionary that define the architecture and data structure of the system.

4.1 Block Diagram

A block diagram provides a high-level graphical representation of the SkillSwap system architecture, showing the interaction between the major components. The system follows a client-server architecture. The React.js frontend communicates with the Express.js backend through REST API calls, and the backend interacts with the MySQL database for data storage and retrieval. All protected routes pass through a JWT authentication middleware layer before reaching the business logic.

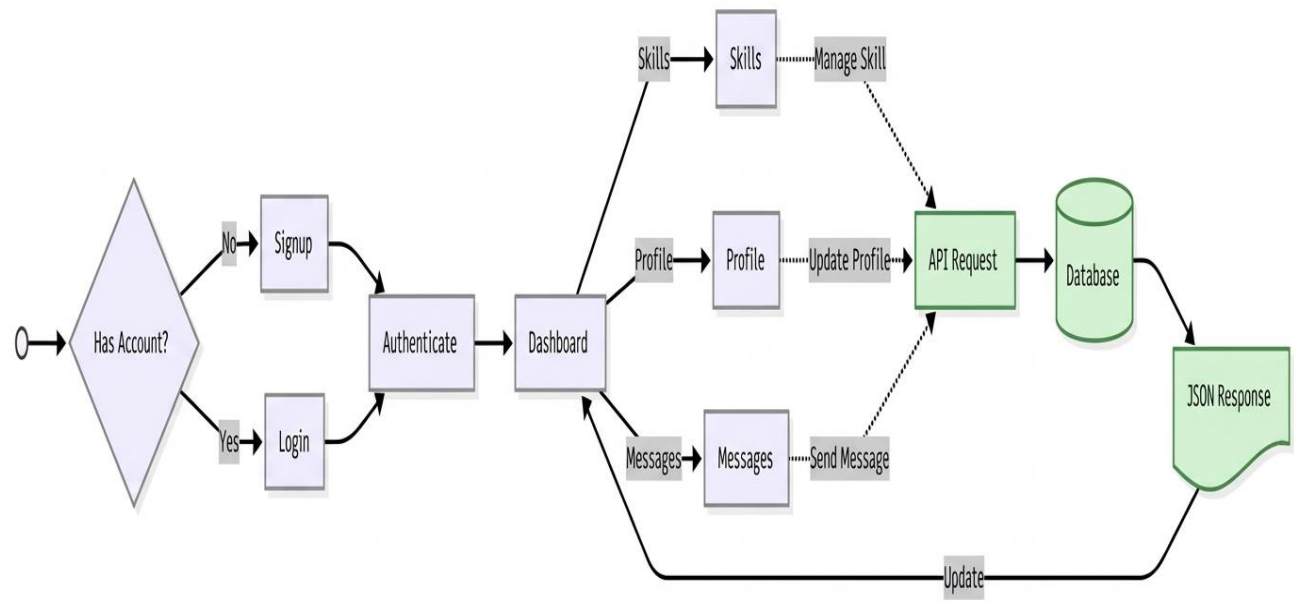

Figure 4.1 – System Block Diagram

4.2 Data Flow Diagrams

The Data Flow Diagram (DFD) illustrates how data moves through the SkillSwap system — from the user interface through the API layer to the database and back. At Level 0, the user interacts with the SkillSwap system which processes requests and returns responses. At Level 1, the system is decomposed into its core processes: Authentication, Skill Management, User Profile Management, and Messaging. Each process exchanges data with the MySQL database and the user interface as appropriate.

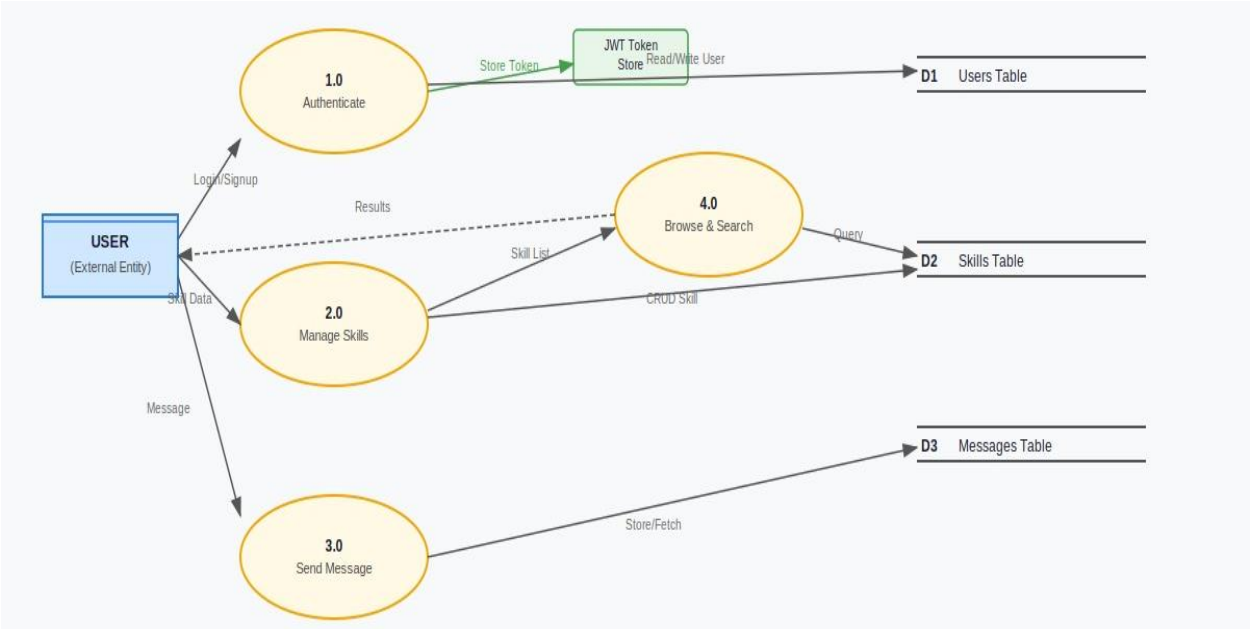

Figure 4.2 – Data Flow Diagram

4.3 ER Diagram

The Entity Relationship Diagram defines the database schema and the relationships between entities in the SkillSwap system. The three primary entities are USERS, SKILLS, and MESSAGES. The relationship between USERS and SKILLS is one-to-many — one user can create many skills, but each skill belongs to exactly one user (referenced by user_id as a foreign key with ON DELETE CASCADE). The relationship between USERS and MESSAGES is one-to-many in two directions — one user can send many messages (sender_id FK) and one user can receive many messages (receiver_id FK).

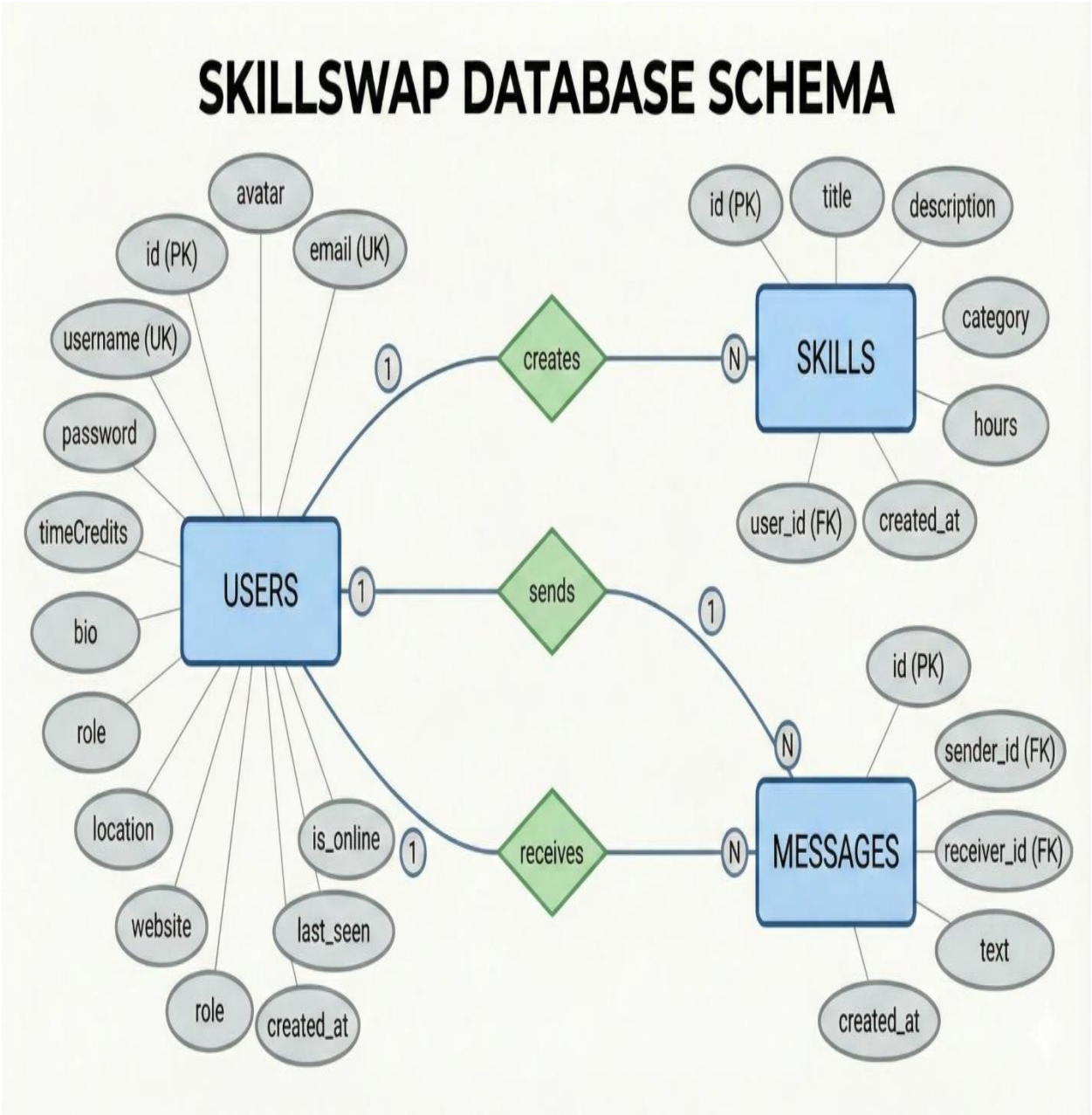

Figure 4.3 – Entity Relationship Diagram

4.4 UML Diagrams

4.4.1 Class Diagram

The Class Diagram represents the static structure of the SkillSwap system, showing the primary classes — User, Skill, Message, and AppContext — along with their attributes, methods, and relationships. The User class contains attributes such as id, username, email, password, timeCredits, and isOnline. The Skill class references a User through a userId foreign key. The Message class references two User objects through senderId and receiverId.

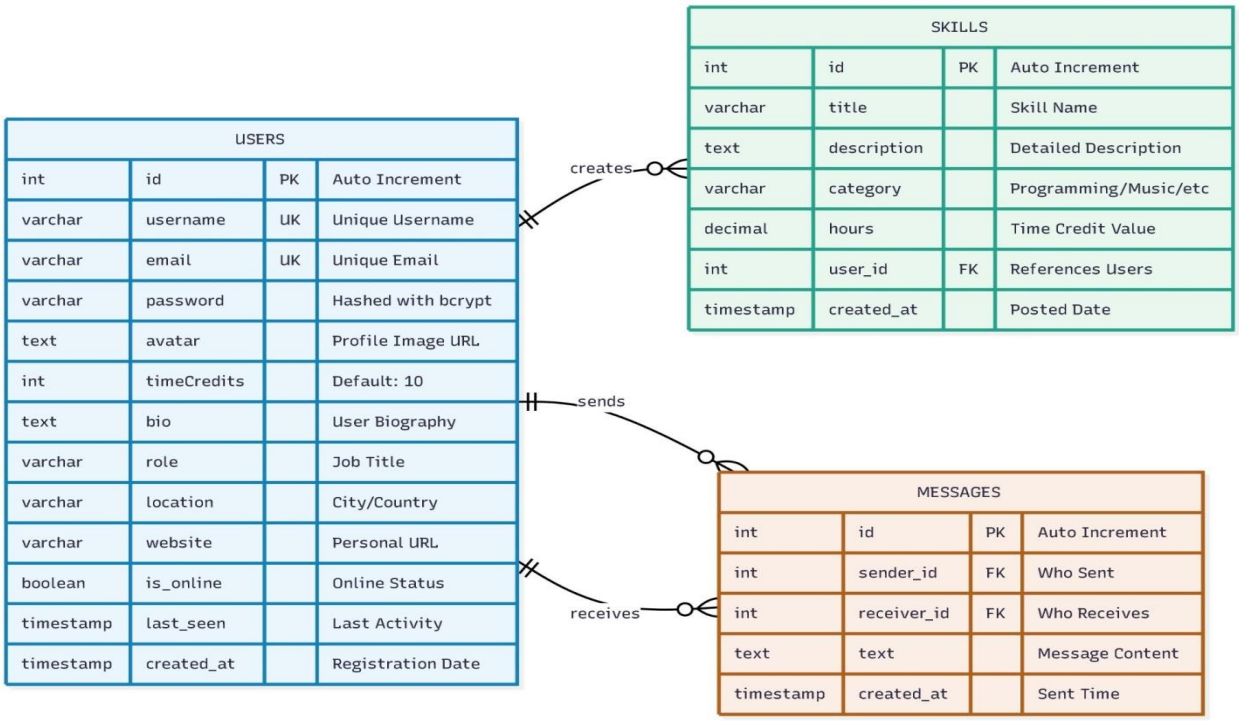

Figure 4.4.1 – Class Diagram

4.4.2 Use Case Diagram

The Use Case Diagram illustrates the interactions between the primary actors — the User and the System — and the core use cases of SkillSwap. The User can Register, Login, Browse Skills, Post a Skill, Send a Message, View and Edit Profile, and Logout. The System handles authentication, token validation, data storage, and status tracking automatically.

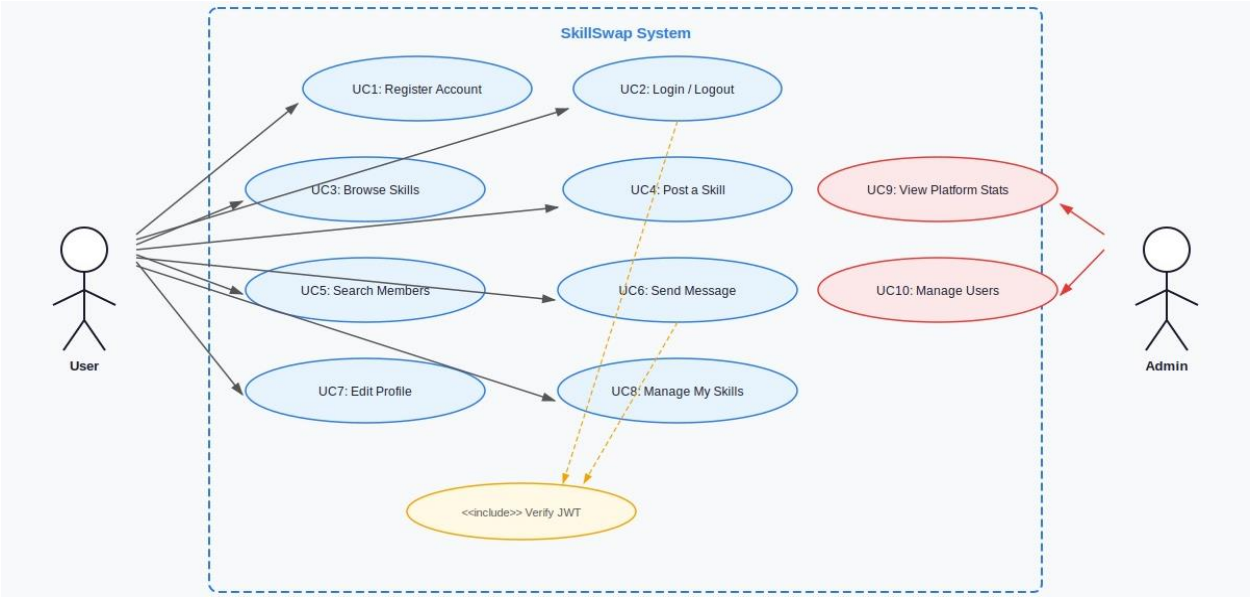

Figure 4.4.2 – Use Case Diagram

4.4.3 Sequence Diagram

The Sequence Diagram for the Authentication flow shows the time-ordered exchange of messages between the User, the React App, the Express API, the JWT Middleware, and the MySQL database. On login, the user submits credentials which the frontend sends to /api/auth/login. The backend retrieves the user record, compares the password using bcrypt, updates the online status to TRUE, generates a JWT token, and returns it to the frontend. The AppContext is updated and the user is redirected to the Dashboard.

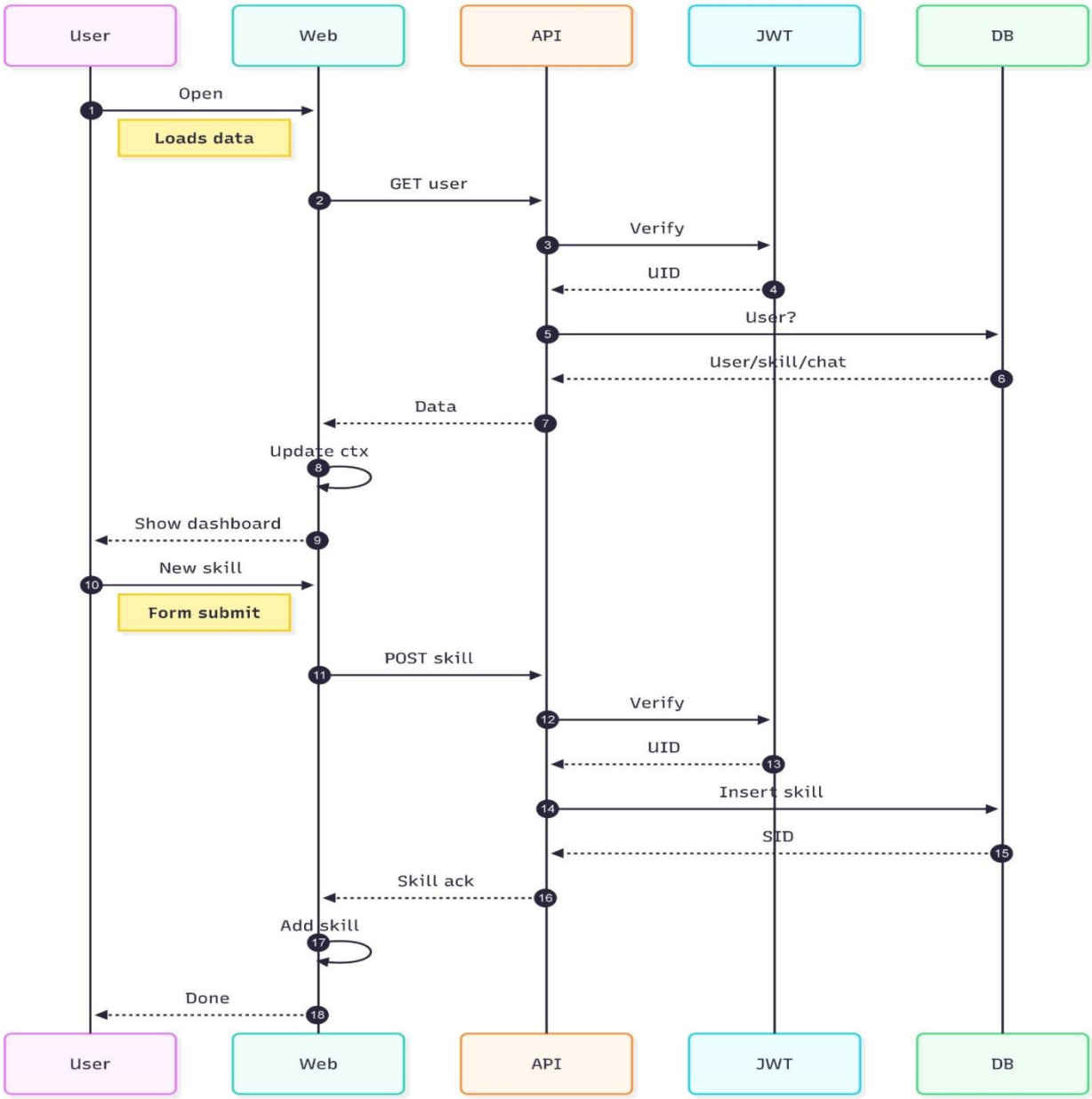

Figure 4.4.3 – Sequence Diagram – Authentication Flow

4.4.4 Collaboration Diagram

The Collaboration Diagram illustrates the structural organization of the objects involved in the Skill Exchange use case and the messages they exchange. The Learner object sends a searchSkills() message to the SkillMarketplace, which queries the SkillsController. The SkillsController retrieves data from the MySQL Database and returns the skills list back to the Marketplace. The Learner then initiates a requestSession() message which flows through the MessagingSystem to the Teacher object.

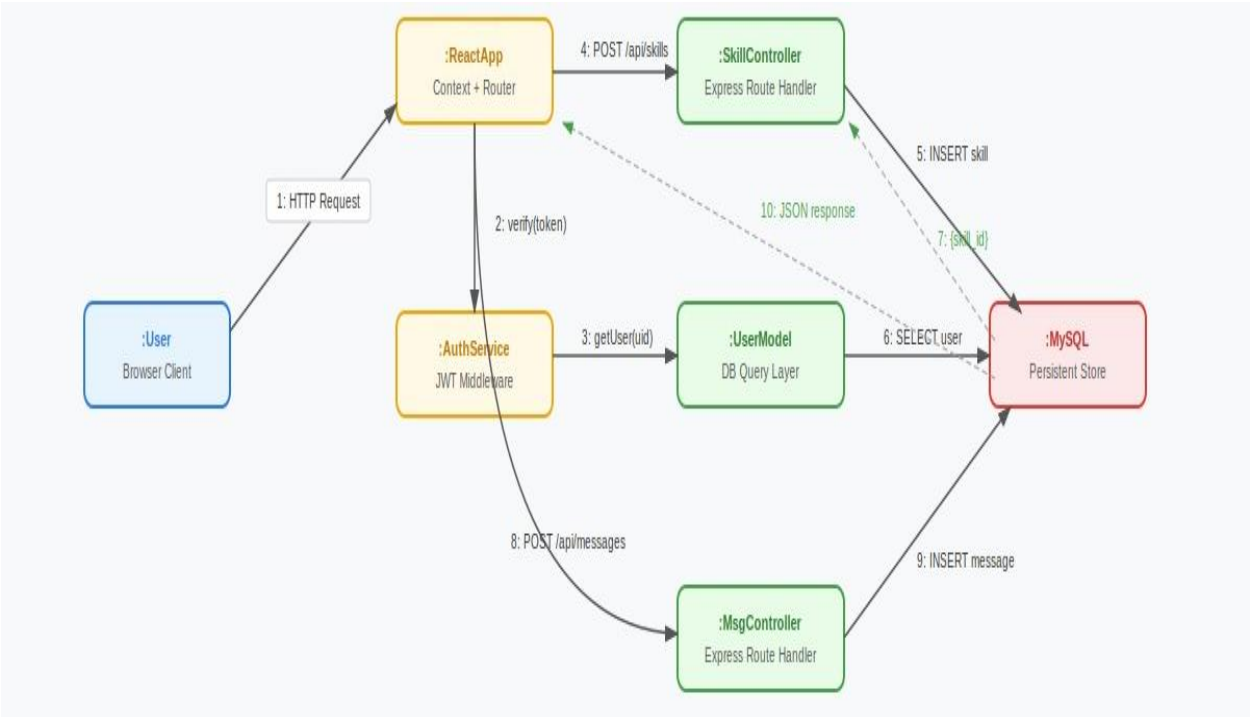

Figure 4.4.4 – Collaboration Diagram

4.4.5 Activity Diagram

The Activity Diagram for the Skill Exchange workflow describes the complete sequence of actions from browsing the marketplace to completing a skill session. The user browses available skills, selects one of interest, views the skill details and teacher profile, initiates a chat, negotiates a session time, conducts the learning session, and upon completion time credits are transferred accordingly between the learner and the teacher.

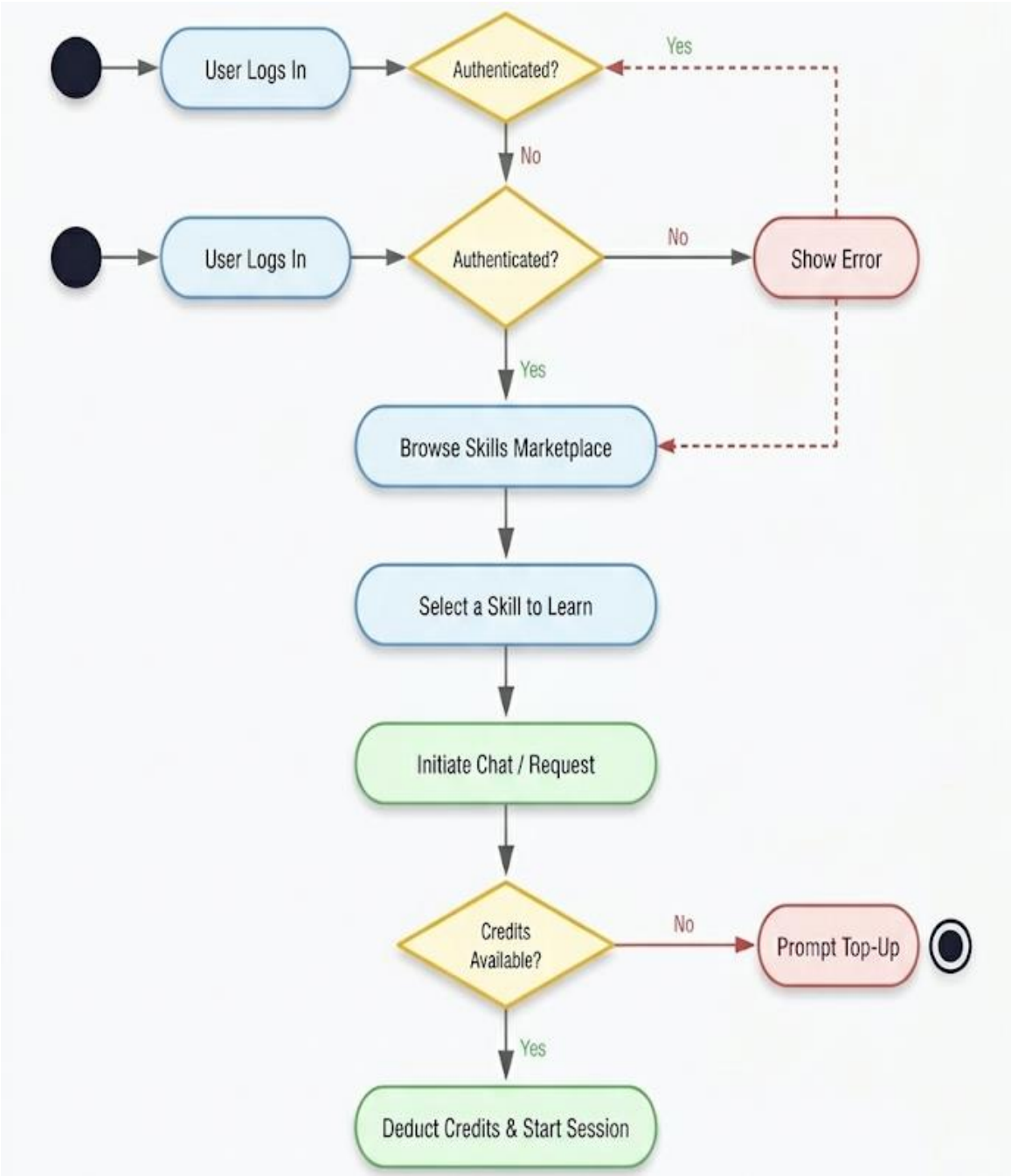

Figure 4.4.5 – Activity Diagram – Skill Exchange

4.5 Data Dictionary

A data dictionary provides a structured description of all the data elements used in the SkillSwap database. It defines each column's name, data type, constraints, and purpose so that developers, testers, and future maintainers clearly understand how data is stored and used across the system. By standardizing field definitions, the data dictionary reduces ambiguity, prevents inconsistent usage of attributes, and makes database changes easier to manage over time .

Table 4.5.1 – Users Table Data Dictionary

Column	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique user identifier
username	VARCHAR	UNIQUE, NOT NULL	Display name of the user
email	VARCHAR	UNIQUE, NOT NULL	User email address
password	VARCHAR	NOT NULL	bcrypt hashed password
avatar	TEXT	–	Profile image URL (DiceBear)
timeCredits	INT	DEFAULT 10	Learning credit balance
bio	TEXT	–	User biography
role	VARCHAR	–	Job title or expertise area
location	VARCHAR	–	City or Country
is_online	BOOLEAN	DEFAULT FALSE	Real-time online status
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp

Table 4.5.2 – Skills Table Data Dictionary

Column	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique skill identifier
title	VARCHAR	NOT NULL	Name of the skill offered
description	TEXT	NOT NULL	Detailed skill description
category	VARCHAR	NOT NULL	Skill category (e.g., Programming)
hours	DECIMAL	–	Time credit value in hours
user_id	INT	FK → Users, CASCADE	References the skill owner
created_at	TIMESTAMP	DEFAULT NOW()	Skill posting timestamp

Table 4.5.3 – Messages Table Data Dictionary

Column	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique message identifier
sender_id	INT	FK → Users, CASCADE	User who sent the message
receiver_id	INT	FK → Users, CASCADE	User who receives the message
text	TEXT	NOT NULL	Message content
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when message was sent

CHAPTER 5

IMPLEMENTATION

5.1 Technology Used

The SkillSwap platform is developed using a modern full-stack JavaScript architecture. Each technology was carefully selected to ensure performance, developer productivity, scalability, and community support. The table below summarises the complete technology stack.

Table 5.1.1 – Technology Stack Used

Layer	Technology	Purpose
Frontend	React 19 + Vite	SPA development with fast Hot Module Replacement
Styling	Tailwind CSS	Utility-first responsive CSS framework
State Mgmt.	React Context API	Global state: user, skills, chat data
Routing	React Router DOM	Client-side navigation between pages
Backend	Node.js + Express.js	RESTful API server with middleware support
Database	MySQL + mysql2	Relational data: users, skills, messages
Auth	JWT + bcrypt	Token-based auth + password hashing
Avatars	DiceBear API	Auto-generated user profile avatars
Deployment	Vercel	Frontend hosting and CI/CD

React 19 with Vite provides a blazing-fast development environment with Hot Module Replacement (HMR). Tailwind CSS enables rapid UI prototyping with a utility-first approach, eliminating the need for custom CSS files. The React Context API manages global application state — including the authenticated user's data, all skills, and conversations — without requiring third-party libraries like Redux.

The backend is built on Node.js with Express.js providing a minimal and flexible RESTful API framework. Authentication is implemented using JWT tokens with 1-hour expiry, verified on every protected route through a dedicated middleware function. Passwords are never stored in plain text — bcrypt with 10 salt rounds generates secure hashes. The MySQL database stores all persistent data across three tables (users, skills, messages), with the mysql2 driver providing promise-based database queries.

5.2 Frontend Design

The frontend of SkillSwap is designed to be modern, responsive, and user-friendly. The application follows Single Page Application (SPA) architecture with React Router DOM handling client-side navigation. All screens share a consistent sidebar navigation component and are styled uniformly using Tailwind CSS utility classes.

5.2.1 Login and Signup Screen

The login and signup screens allow new users to create accounts and existing users to access the platform. The signup form collects username, email, and password. Upon submission, a POST request is sent to /api/auth/signup. The backend hashes the password using bcrypt with 10 salt rounds, stores the user record in MySQL, generates a JWT token, and returns it to the frontend. The token is stored in localStorage and the user is redirected to the Dashboard.

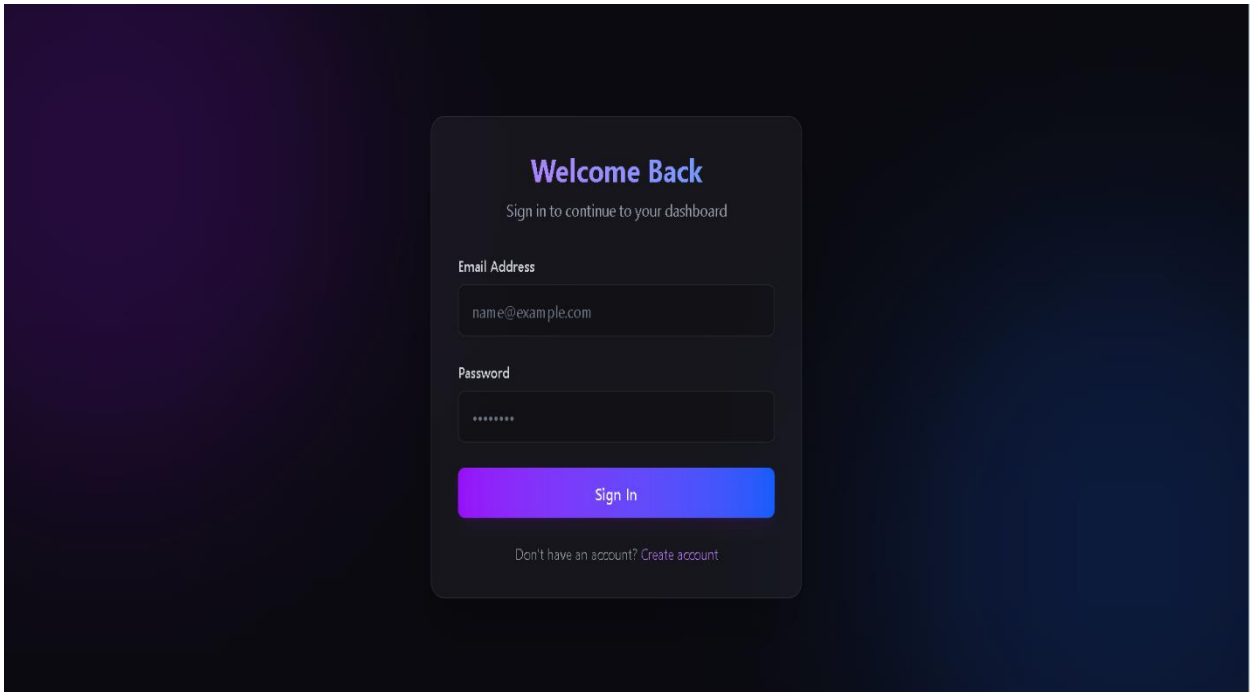

Figure 5.2.1 – Login / Signup Screen

5.2.2 Dashboard Screen

The Dashboard is the central navigation hub of the SkillSwap application. It displays community statistics including total registered members, skills listed, and currently online users. A featured skill grid presents recently posted skills, and a member discovery

section highlights new community members. Quick-access buttons navigate to Browse Skills and Browse Members pages.

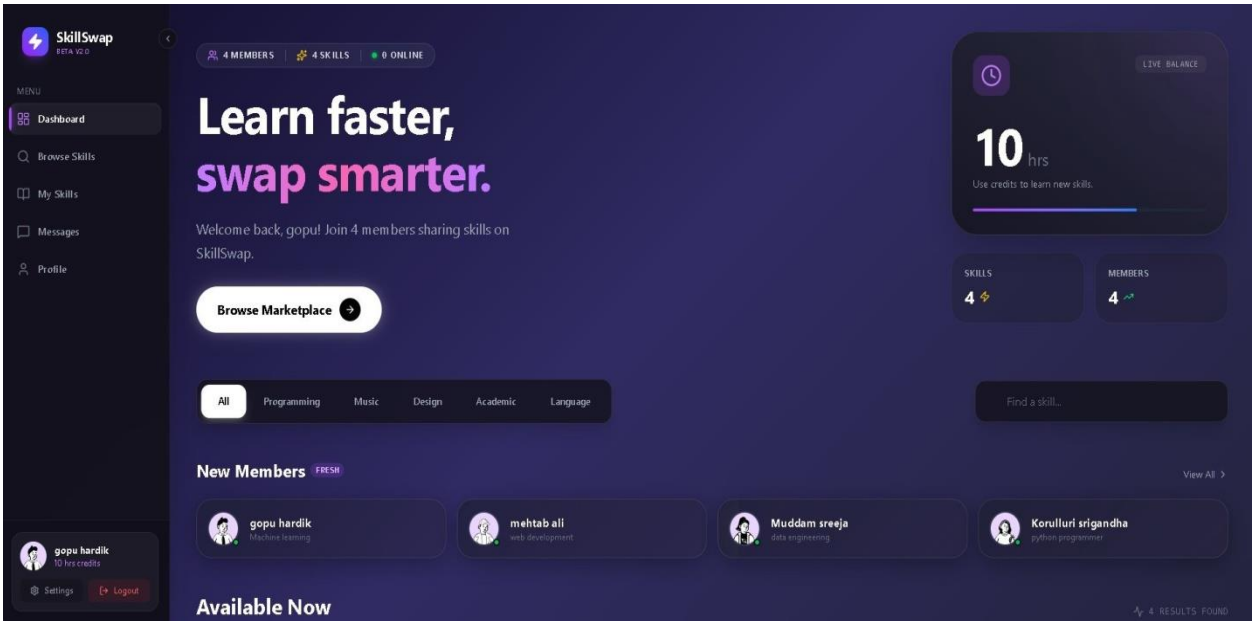

Figure 5.2.2 – Dashboard Screen

5.2.3 Browse Skills Marketplace

The Browse Skills page displays all available skill listings from community members. Users can filter skills by category using filter buttons (All, Programming, Music, Design, Business, Language, etc.) and search for specific skills. Each skill card displays the title, description, category tag, time-credit value, the offering user's name and online status, and action buttons to chat or start a session.

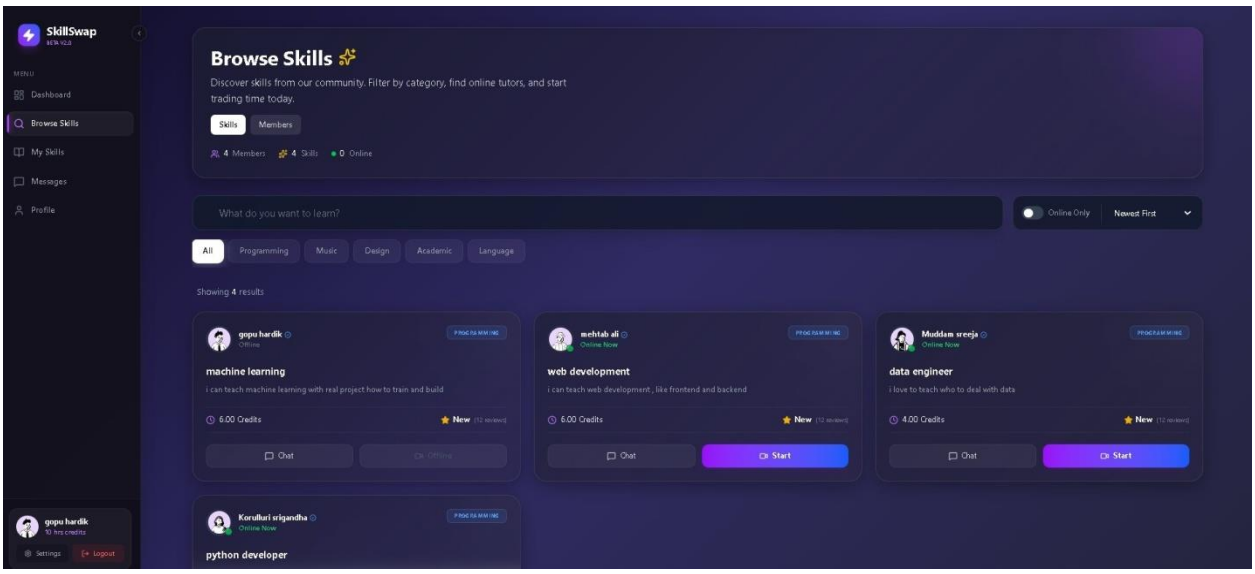

Figure 5.2.3 – Browse Skills Marketplace

5.2.4 Browse Community Members

The Members page displays all registered users on the platform with their profile picture, username, role, bio, and online/offline status. A search bar allows users to find members by name. This page enables users to discover potential learning partners and directly initiate conversations without going through the skill marketplace.

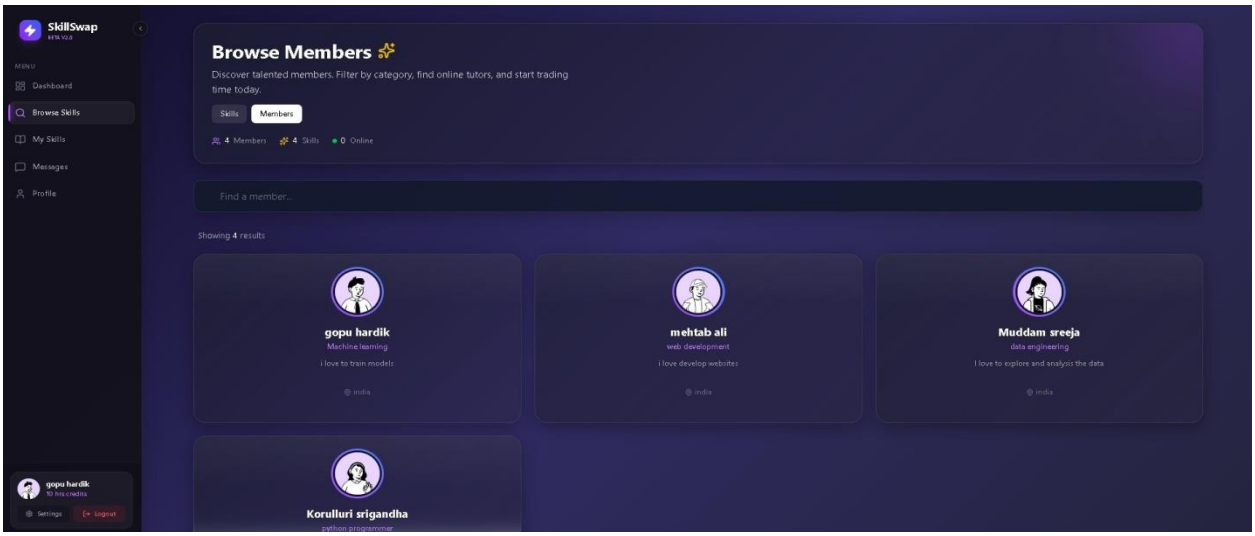

Figure 5.2.4 – Browse Community Members

5.2.5 My Skills Management

The My Skills page provides each user with a personal dashboard for managing their skill portfolio. Users can view their listed skills, number of skills posted, time credits earned, and pending requests. An 'Add New Skill' button opens a form to create a new skill listing. Each listed skill card provides options to edit or delete the skill, giving users full control over their marketplace presence.

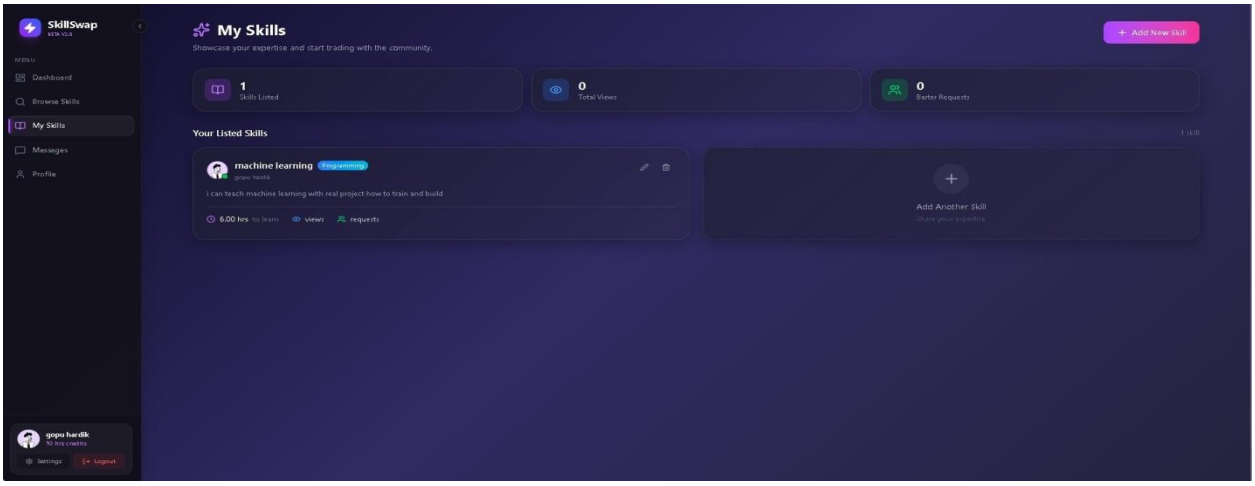

Figure 5.2.5 – My Skills Management

5.2.6 User Profile Screen

The User Profile page displays comprehensive information about a user, including their avatar, username, role, bio, location, website, time-credit balance, skills listed count, and profile views. Users can edit their own profile through an Edit Profile button. The profile page also shows the user's listed skills so visitors can quickly decide whether to initiate a learning session.

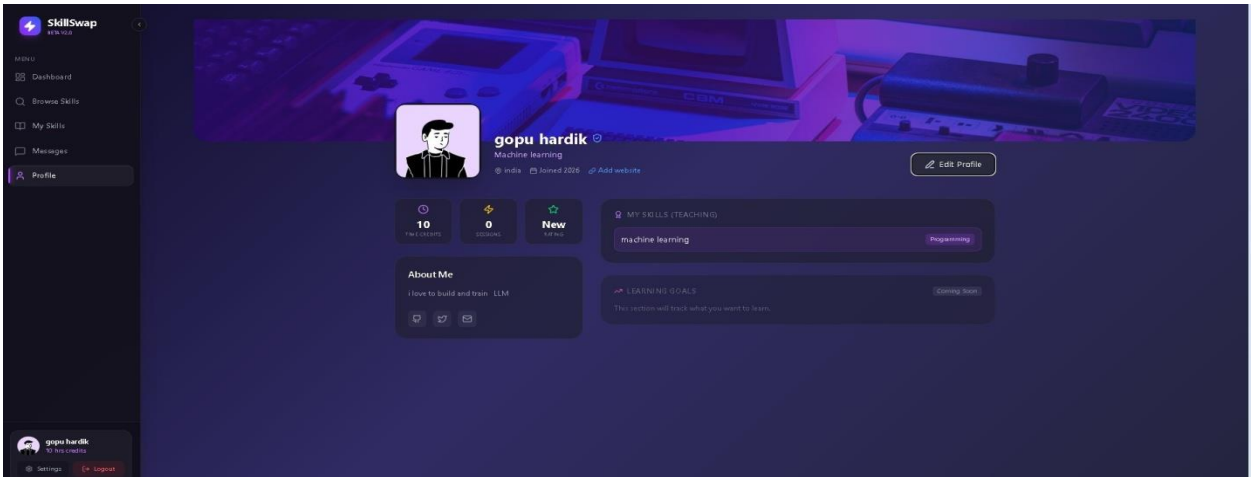

Figure 5.2.6 – User Profile Screen

5.2.7 Messaging System

The Messages page provides a real-time chat interface for communication between users. The left panel lists all active and recent conversations with the contact's name, avatar, and last message preview. The right panel displays the full message thread for the selected conversation with timestamps. Users type and send messages using the input field at the bottom. All messages are persisted in the MySQL database and retrieved on page load.

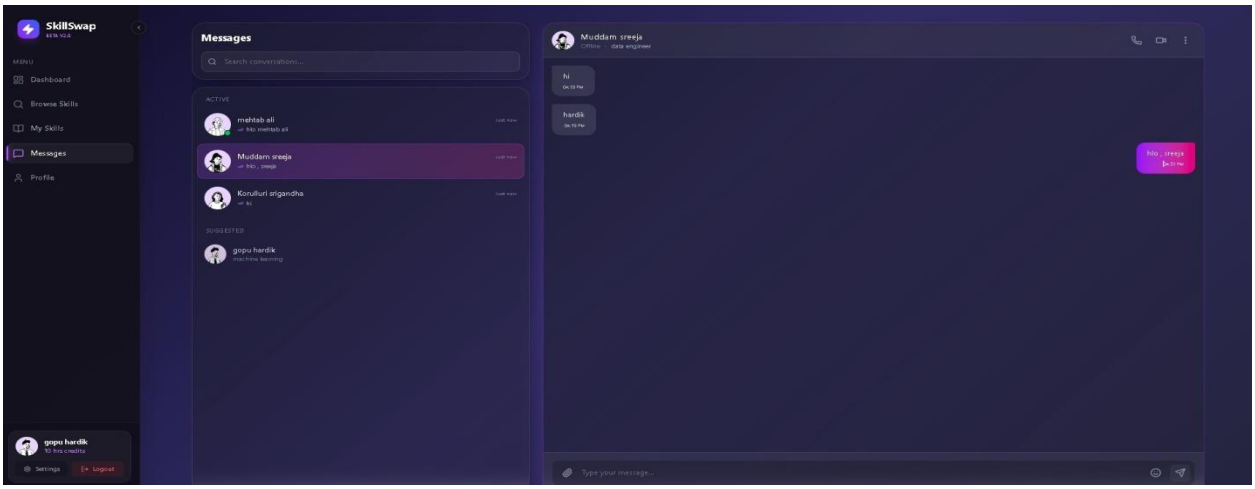

Figure 5.2.7 – Messaging System

CHAPTER 6

TESTING

Testing is a critical phase of software development that ensures the SkillSwap application functions correctly and meets all specified requirements. This chapter describes the testing strategy, the test cases executed for each module, the problems encountered, and their counter measures.

6.1 Testing Strategy

- API Testing: Individual REST API endpoints were tested using Postman to verify correct responses, HTTP status codes, and data formats for both valid and invalid inputs.
- Integration Testing: The correctness of data flow between the React frontend and the Express.js backend was verified — ensuring that data displayed in the UI accurately reflects records stored in MySQL.
- Cross-Browser Testing: The UI was tested on Google Chrome, Microsoft Edge, and Mozilla Firefox to ensure consistent appearance and functionality across browsers.
- Database Testing: Data persistence was verified in MySQL Workbench by inspecting records after user registration, skill creation, and message exchange operations.
- End-to-End Testing: Complete user flows from registration through skill posting, browsing, and messaging were tested as integrated workflows to simulate real user behaviour.

6.2 Test Cases and Results

Table 6.1 – Test Cases and Results Summary

No.	Module	Test Description	Expected Result	Actual	Status
1	Registration	Register with valid details	User stored in DB, JWT returned	As expected	PASS
2	Registration	Duplicate email registration	Error: Email already exists	As expected	PASS
3	Login	Login with correct credentials	JWT returned, redirect to Dashboard	As expected	PASS

4	Login	Login with wrong password	Error: Invalid credentials	As expected	PASS
5	Skill Post	Create skill with all fields	Skill stored and shown on browse page	As expected	PASS
6	Skill Filter	Filter by Programming category	Only programming skills displayed	As expected	PASS
7	Messaging	Send message between two users	Stored in DB and shown in chat	As expected	PASS
8	Messaging	Reload messages page	Previous messages still visible	As expected	PASS
9	Status	Login and logout tracking	is_online toggles correctly in DB	As expected	PASS
10	Security	Access protected route without token	401 Unauthorized returned	As expected	PASS
11	Navigation	Navigate between all pages	Smooth transitions, correct data	As expected	PASS
12	Responsive	Test on mobile screen size	Layout adapts for mobile correctly	As expected	PASS

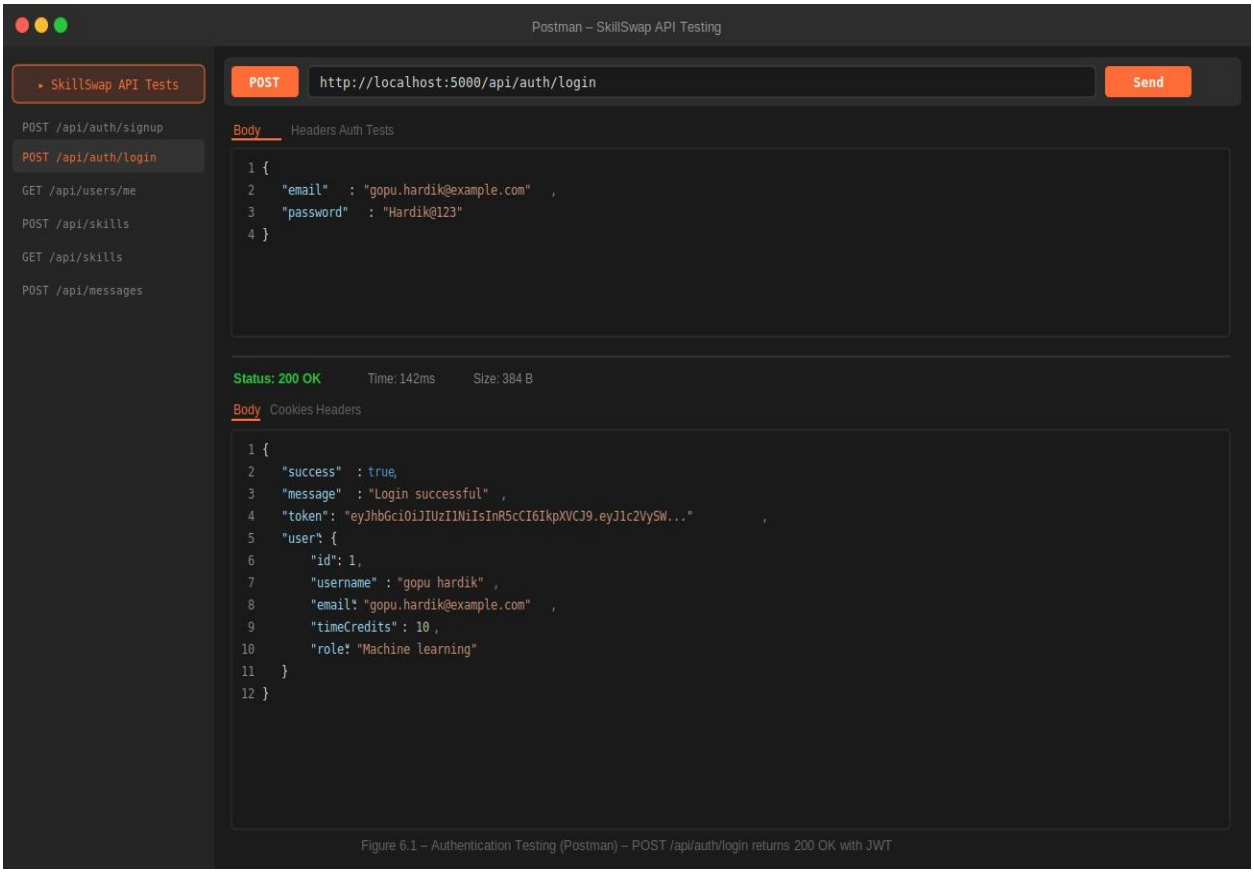

Figure 6.1 – Authentication Testing Screenshot

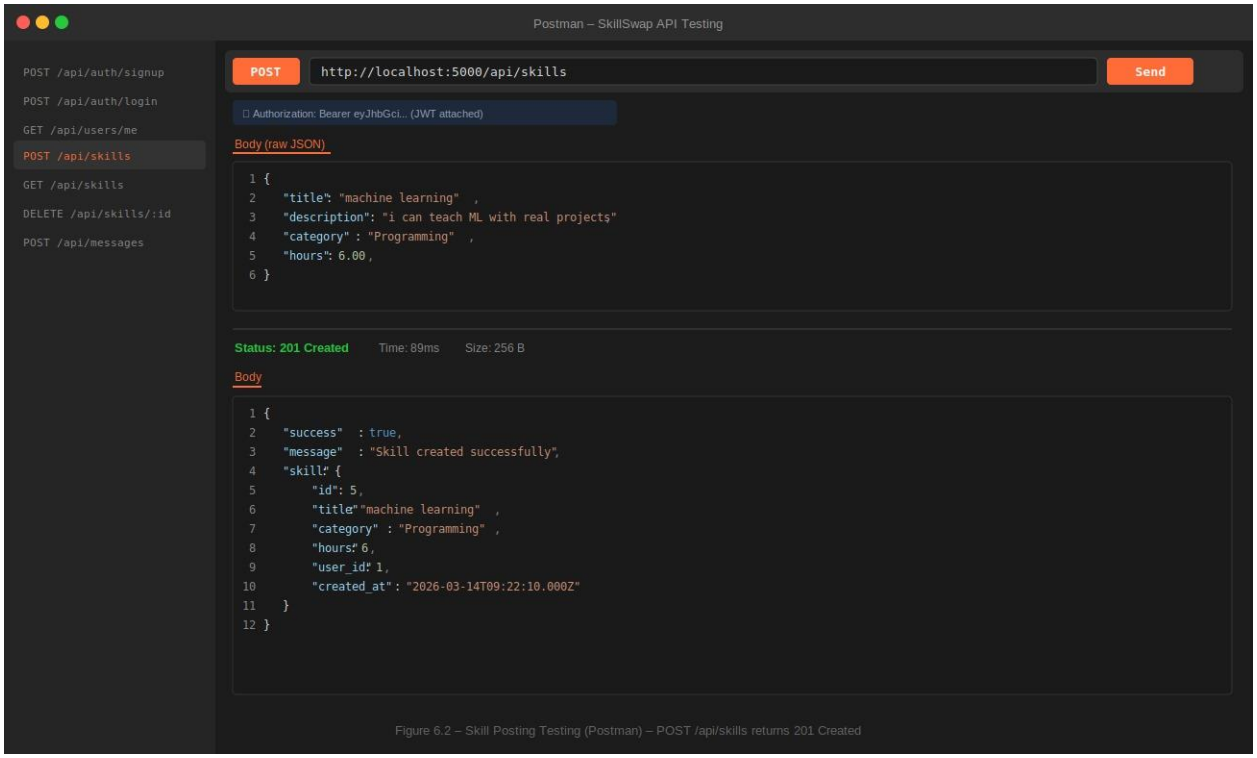

Figure 6.2 – Skill Posting Testing Screenshot

6.3 Problems Encountered and Counter Measures

Table 6.2 – Feature Implementation Status

Feature	Status
User Authentication (Register / Login / Logout)	Complete
Skill CRUD Operations (Create / Edit / Delete)	Complete
Browse and Filter Skills Marketplace	Complete
Real-time Chat Messaging with Persistence	Complete
User Profile Management	Complete
Online Status Tracking (is_online)	Complete
Community Dashboard with Statistics	Complete
Responsive UI (Mobile and Desktop)	Complete
Video Calls (WebRTC)	UI Ready – WebRTC Integration Pending
Rating and Review System	Planned

CHAPTER 7

RESULTS

This chapter presents the results obtained after successful development, deployment, and testing of the SkillSwap application. The system was tested with real user data stored in the MySQL database, and all functional modules were verified to perform as expected. Sample screenshots of the project are presented below along with explanations.

7.1 Platform Statistics Overview

During testing, four users were registered on the platform and actively used its features. The MySQL database recorded the following platform statistics:

- Total Registered Users: 4
- Total Skills Listed: 4 (all Programming category)
- Total Messages Exchanged: 7
- Platform deployed and accessible live at: https://skillswap1-ashen.vercel.app

7.2 Registered Users Data

The following users were registered during testing, demonstrating correct data storage, JWT authentication, and profile creation in the MySQL database:

- User 1: Gopu Hardik – Machine Learning (India) – Time Credits: 10
- User 2: Mehtab Ali – Web Development (India) – Time Credits: 10
- User 3: Muddam Sreeja – Data Engineering (India) – Time Credits: 10
- User 4: Kolluri Srigandha – Python Developer (India) – Time Credits: 10

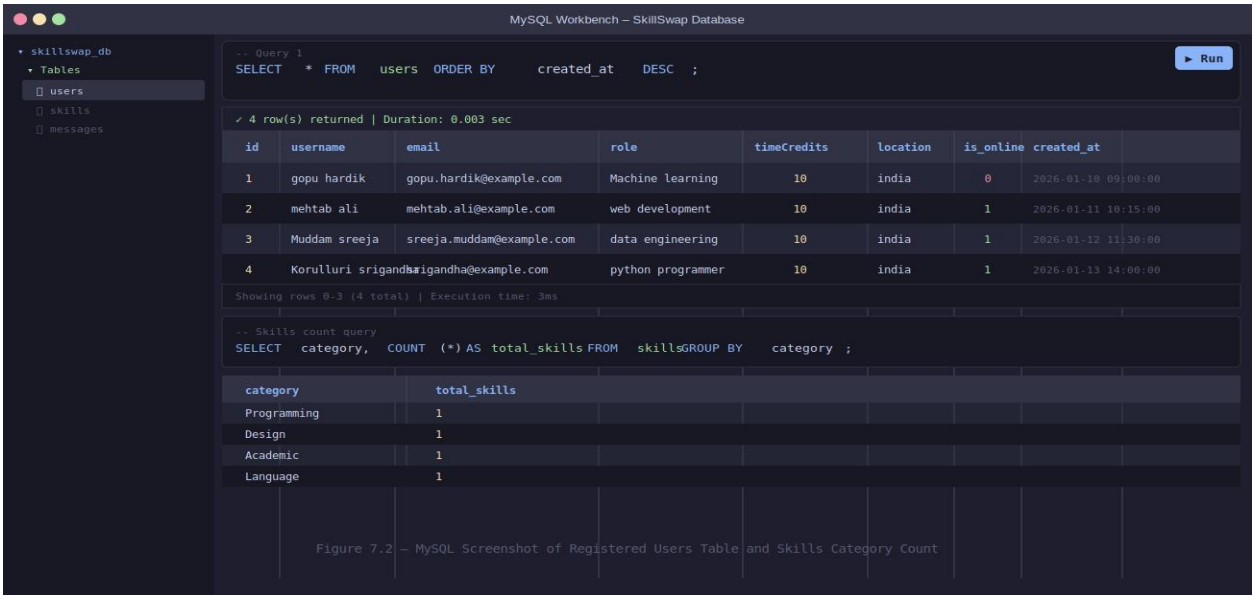

Figure 7.1 – Registered Users Data in MySQL Database

7.3 Skills Marketplace Data

Four skills were posted during testing, each correctly stored in the MySQL skills table with the creator's user_id and time-credit value:

- Skill 1: Machine Learning – Programming – 6 hrs – User 1 (Gopu Hardik)
- Skill 2: Web Development – Programming – 6 hrs – User 2 (Mehtab Ali)
- Skill 3: Data Engineering – Programming – 4 hrs – User 3 (Muddam Sreeja)
- Skill 4: Python Developer – Programming – 3 hrs – User 4 (Kolluri Srigandha)

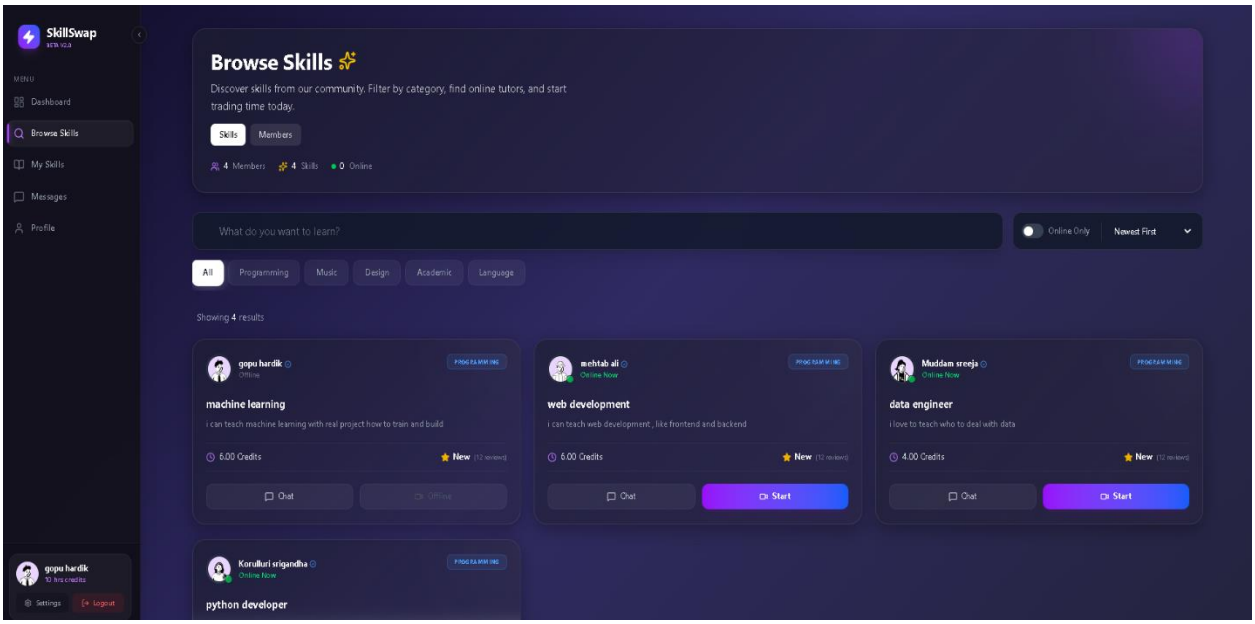

Figure 7.2 – Skills Marketplace Data

CHAPTER 8

CONCLUSION

The SkillSwap application was successfully designed, developed, and tested as a Real-Time Resource Pool (RTRP) project for the academic year 2025–26. The platform demonstrates a practical and effective implementation of the time-banking concept in a modern web environment, enabling peer-to-peer skill exchange without monetary transactions.

The project achieves its primary objective of creating an accessible, community-driven learning platform where time is the currency of exchange. Users can register, create profiles, list their skills, browse and discover skills offered by others, and communicate directly through a real-time messaging system. The application is built on a robust full-stack architecture using React.js, Node.js, Express.js, and MySQL, with JWT-based security ensuring user data protection at all times.

The development process provided the team with comprehensive practical experience in full-stack web development — from designing REST APIs and implementing relational database schemas to building responsive single-page applications and handling secure authentication flows. The project successfully demonstrates how modern web technologies can be combined to solve real-world social and educational problems with no monetary barrier.

Testing confirmed that all functional modules — authentication, skill management, community browsing, messaging, and online status tracking — operate correctly and reliably. The platform was deployed on Vercel, making it accessible as a live web application. Overall, SkillSwap successfully serves as a functional proof-of-concept for a demonetized, community-driven educational exchange platform, with a strong architectural foundation ready for future production-level scaling.

FUTURE SCOPE AND ENHANCEMENTS

The SkillSwap platform has been built as a strong foundation for a fully-featured, production-ready peer-to-peer learning ecosystem. Several significant enhancements are planned to increase its functionality, scalability, and social impact.

9.1 Video Call Integration (WebRTC)

WebRTC-based in-app video conferencing will be integrated to enable live skill teaching sessions directly within the platform. The video call interface UI has already been scaffolded; the WebRTC signaling server and peer connection logic are planned for the next development sprint.

9.2 Rating and Review System

A rating and review system will allow learners to rate teachers on a 1–5 star scale with written comments after each session. The teacher's average rating will be displayed on their profile and skill cards, helping new users identify high-quality teachers and improving overall platform quality.

9.3 AI-Powered Skill Matching

An AI-based recommendation algorithm will suggest skills and learning partners based on a user's profile, interests, learning history, and browsing patterns. This will significantly improve the user experience and session success rate.

9.4 Push Notification System

A push notification system will alert users in real time about new messages, skill requests, session reminders, and platform activity. Browser-based push notifications will ensure users never miss important interactions.

9.5 Mobile Application (React Native)

A React Native mobile application for iOS and Android will extend SkillSwap's reach to smartphone users, sharing the same backend API for consistent data and functionality across platforms.

9.6 Cloud Database and Scalability

The MySQL instance will be migrated to a cloud-based database service (PlanetScale, AWS RDS, or Google Cloud SQL) to enable horizontal scaling, automated backups, and global availability for a larger user base.

Table 9.1 – Challenges Faced and Solutions

Challenge	Solution Applied
Database migration from SQLite to MySQL	Used mysql2 driver with environment variables for config
Global state synchronization across pages	Implemented React Context API for user, skills, chats
Protecting API routes from unauthorized access	JWT authentication middleware on all protected endpoints
Real-time online status tracking	Added is_online column toggled on login/logout events
Responsive layout across screen sizes	Used Tailwind CSS Grid and Flexbox utility classes
Persistent chat message history	Stored messages in MySQL with sender/receiver FK references

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